AUTUMN MID SEMESTER EXAMINATION-2023

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Design and Analysis of Algorithms
[CS2012]

Time: 1 1/2 Hours **Full Mark: 40**

Answer Any four Questions including Question No. 1 which is compulsory.
The figures in the margin indicate full marks. Candidates are required to give theiranswers in their own words as far as practicable and all parts of a question should beanswered at one place only.

1. Answer all the questions. [2 x 5 = 10]

a)

I. What is the time complexity of the following function, fun()?

```
int fun(int n){
    int i, j, s=0;
    for(i= 2; i<n; i=i+1)
       for(j = 1; j < n; j= j * i)
              s=s+j;
    return s;
}
```

A. Θ(n) B. Θ(n*log(n)) C. Θ(n^2) D. Θ(n√n) E. NONE

II. What exact value will the function fun() return if the value of 'n' is 5?

b) You are executing an algorithm with worst-case time complexity $O(n^2)$ on a CPU that can perform 10^6 operations per second. Calculate the time required to solve a worst-case input of size 100.

c)

I. Solve the recurrence using the master theorem.
$T(n) = 16T(n/4) + n$

II. In a binary max-heap containing 'n' number, the smallest element can be found in time
A. O(n) B. O(log n) C. O(log log n) D. O(1)

d) Match the following algorithms with their recurrences:

Bubble-sort	$T(n) = T(n/2) + \Theta(1)$
Quick-sort	$T(n) = T(n-1) + \Theta(n)$
Merge-sort	$T(n) = T(k) + T(n-k) + \Theta(n)$
Binary-search	$T(n) = 2T(n/2) + \Theta(n)$

e) Find the time complexity of the following C function, assume n>0.

```
int recursive(int n){
    if(n==1)
        return 1;
    else
        return (recursive(n-1)+recursive(n-1));
}
```

2. Solve the following recurrences: [5 x 2 = 10]

a) $T(n) = 2\,T(\sqrt{n}) + \log n$ and $T(1) = 1$

b) $T(n) = T(n/10) + T(9n/10) + n$

3. [5 x 2 = 10]

a) Write a recursive algorithm to perform a linear search that searches for the given element in the first location and, if not found, it recursively calls the linear search. Derive the recurrence relation for recursive linear search and solve the recurrence using the iterative method.

b) Consider an alternative to binary search called ternary search, which divides the input array into three equal parts and recursively searches one of these three segments. Write the algorithm and find the key element 13 from the array A = {2, 3, 5, 6, 8, 9, 12, 13, 14} with indices from 1 to 9 using the above procedure.

4. [5 x 2 = 10]

a) Write down the PARTITION(A, p, r) procedure with the last element as the pivot, where p and r are lower & upper bounds of array A. Describe in a step-by-step process how to get the pass1 result of PARTITION() by taking the last element as the pivot on the following array elements. Derive the time complexity of the PARTITION() procedure.

A = {2, 5, 7, 5, 9, 3, 8, 6}

b) Discuss the trade-offs between the Quick-Sort and Merge-Sort algorithms. Compare and contrast their time complexity, space complexity, and stability.

5. [5 x 2 = 10]

a) Write the algorithm to build a MIN-HEAP. Describe your Algorithm to build a MIN-HEAP from the array A = {5, 7, 8, 2, 1, 0, 3, 9, 4, 5, 6} in a step-by-step process. Derive the time complexity of building a MIN-HEAP. Assume the root is at index 1.

b) Given a knapsack with a capacity of 15 kg and a set of items with their weights in kg and profits in rupees:
Items: A B C
Weights: 8 4 10
Profits: 30 15 45
Solve the fractional knapsack problem to maximize the total profits obtained in each of the following cases and find the case(s) with highest profit.

I. Greedy w.r.t. profits only
II. Greedy w.r.t. weights only
III. Greedy w.r.t. profit per unit weight

*** Best of Luck ***